Supplementary Information

Graph-inverted Ghanaian aquifers under aridification

This Supplementary Information provides methodological detail, data descriptors, quality-control summaries, graph-construction assumptions, inverse-modelling logic, uncertainty treatment and reproducibility information required to support the main manuscript. The supporting dataset workbook contains 160 borehole nodes, 320 wet/dry hydrochemical samples, 140 regional climate records, 397 borehole-level graph edges and 320 model-feature records.

Supplementary Methods

Supplementary Method 1. Study design and sampling frame

The study domain comprises the Northern, North East, Upper East and Upper West regions of Ghana. The dataset is balanced by region, with 40 boreholes per region and two seasonal groundwater samples per borehole, yielding 320 hydrochemical records. The boreholes span Middle Voltaian sedimentary aquifers, Birimian fractured basement aquifers, granitoid fractured basement aquifers and regolith/alluvial shallow aquifers. This design supports comparison across lithological, climatic, hydrochemical and land-use gradients while preserving a borehole-level graph representation of the groundwater system. Borehole attributes include coordinates, elevation, borehole depth, static water level, aquifer type, geology group, lithology, land use, distances to rivers, farms and settlements, recharge-zone score, aridity-stress score, fluoride-geogenic-risk score and anthropogenic-risk score. Exact coordinates are spatially masked in public data releases where community water-supply security or ethical concerns apply.

Supplementary Table 1. Workbook structure and record counts.

Workbook sheet	Records	Content
Dataset_Info	1	Study metadata, purpose and dataset design
Dashboard	1	Headline counts and regional summary
Wells_Nodes	160	Borehole-level graph nodes, geometry, aquifer type, land use and risk scores
Hydrochemistry_Seasonal	320	Wet- and dry-season groundwater chemistry and charge-balance information
Climate_Aridification	140	Regional annual climate and vegetation-stress variables for 1991–2025
Graph_Edges	397	Borehole-level graph edges and edge-weight components
Model_Feature_Matrix	320	Model-ready process scores and aquifer-evolution labels
Data_Dictionary	19	Variable definitions
Quality_Control	8	Structural consistency and QA/QC summary

Supplementary Table 2. Regional sampling balance and dominant aquifer classes.

Region	Boreholes	Seasonal samples	Dominant aquifer classes
North East Region	40	80	Birimian (11), Middle (22), Regolith/alluvial (7)
Northern Region	40	80	Birimian (8), Middle (26), Regolith/alluvial (6)
Upper East Region	40	80	Birimian (10), Granitoid (14), Regolith/alluvial (16)
Upper West Region	40	80	Birimian (14), Granitoid (10), Middle (7), Regolith/alluvial (9)

Supplementary Table 3. Core sampling, climate, graph and quality-control counts.

Item	Value	Interpretation
Boreholes	160	one graph node per borehole
Seasonal hydrochemical samples	320	wet and dry samples per borehole
Wet-season samples	160	one per borehole
Dry-season samples	160	one per borehole
Climate records	140	four regions × 35 years (1991–2025)
Graph edges	397	borehole-level weighted edges
Model-feature records	320	one model-feature row per seasonal sample
Samples within ±5% CBE	91.9%	294 of 320 samples
Fluoride > 1.5 mg L^{-1}	43	13 wet-season and 30 dry-season samples
Nitrate > 50 mg L^{-1}	51	22 wet-season and 29 dry-season samples

Supplementary Table 4. Borehole metadata summary.

Borehole attribute	Minimum	Mean	Maximum
Latitude	8.471	10.206	11.121
Longitude	-3.171	-1.134	0.172
Elevation_m	141.000	262.709	340.400
Borehole_Depth_m	20.000	53.629	99.900
Static_Water_Level_m	7.800	16.549	35.000
Distance_River_km	0.100	6.370	25.020
Distance_Farm_km	0.050	1.959	9.970
Distance_Settlement_km	0.050	2.279	10.020
Recharge_Zone_Score_0_1	0.050	0.335	0.748
Aridity_Stress_0_1	0.406	0.662	0.875
Fluoride_Geogenic_Risk_0_1	0.020	0.359	0.970
Anthropogenic_Risk_0_1	0.020	0.439	0.852

Note: Coordinates are summarized only as ranges; public data releases use spatial masking where community water-supply security or ethical considerations require it.

Supplementary Method 2. Field sampling and sample handling

Each borehole was represented as a graph node and sampled in wet and dry seasons. Field sampling followed a consistent borehole-purging protocol until pH, electrical conductivity and temperature stabilized, defined by three consecutive readings with less than 5% variation. Field records include pump type, sampling date, borehole condition, land-use setting, distance to potential contamination sources, and notes on turbidity or sampling anomalies. The workbook records one wet-season and one dry-season sample for each borehole.

Supplementary Method 3. Laboratory analysis and analytical QA/QC

The hydrochemical dataset includes pH, EC, TDS, temperature, Ca^{2+}, Mg^{2+}, Na^{+}, K^{+}, HCO_3^{-}, Cl^{-}, SO_4^{2-}, NO_3^{-}, F^{-}, $\delta^{18}O$, $\delta^{2}H$, Sr, SiO_2, saturation indices, cation and anion milliequivalent sums, charge-balance error, SAR and sodium percentage. Laboratory reporting limits, instrument models, calibration criteria, certified-reference-material recoveries, duplicate precision and blank outcomes are reported in Supplementary Tables 14-18. Major-ion quality was screened using charge-balance error. In the workbook, 294 of 320 samples, or 91.9%, fall within ±5% charge-balance error. Records outside ±5% were reviewed against analytical quality-control criteria and excluded from quantitative inverse-pathway calculations where imbalance remained persistent.

Supplementary Table 5. Summary statistics for measured and derived hydrochemical variables.

Parameter	n	min	Mean	median	max	std
pH	320	6.51	7.11	7.1	7.89	0.265
EC ($\mu S\ cm^{-1}$)	320	427	723.186	718.35	1192.2	151.841
TDS ($mg\ L^{-1}$)	320	289.5	456.445	460.85	719.6	91.014
Temperature (°C)	320	27	29.214	29.2	31.5	0.835
Ca^{2+} ($mg\ L^{-1}$)	320	24.54	41.639	39.94	70.41	10.19
Mg^{2+} ($mg\ L^{-1}$)	320	6.79	15.173	14.52	26.06	4.163
Na^{+} ($mg\ L^{-1}$)	320	15.49	46.002	43.795	84.65	13.081
K^{+} ($mg\ L^{-1}$)	320	2.56	5.351	5.07	10.04	1.456
HCO_3^{-} ($mg\ L^{-1}$)	320	81.59	153.575	149.88	253.09	34.614
Cl^{-} ($mg\ L^{-1}$)	320	20.43	60.157	58.265	124.36	20.199
SO_4^{2-} ($mg\ L^{-1}$)	320	18.48	40.986	38.665	82.18	12.294
NO_3^{-} ($mg\ L^{-1}$)	320	0.57	31.045	24.06	160.46	22.175
F^{-} ($mg\ L^{-1}$)	320	0.04	0.886	0.66	3.88	0.675
$\delta^{18}O$ (‰)	320	-4.96	-3.68	-3.65	-2.45	0.51
$\delta^{2}H$ (‰)	320	-31.95	-23.397	-23.36	-15.04	3.377
Sr ($mg\ L^{-1}$)	320	0.072	0.203	0.202	0.329	0.048
SiO_2 ($mg\ L^{-1}$)	320	21.8	53.258	53.8	81.7	10.671
SAR	320	0.619	1.546	1.54	2.572	0.329
Na (%)	320	24.359	38.971	38.84	50.142	4.498

Note: Detection limits, reporting limits and instrument models are listed in Supplementary Table 14.

Supplementary Method 4. Climate and remote-sensing processing

Climate forcing was summarized annually for 1991–2025 across the four northern Ghana regions. Variables include rainfall, rainfall anomaly, mean temperature, potential evapotranspiration, aridity index P/PET, 12-month SPEI, NDVI, recharge proxy and drought class. The aridity index was calculated as AI = P/PET, where P is annual precipitation and PET is potential evapotranspiration. Declining AI indicates increasing aridification. The climate table contains 140 records, corresponding to four regions across 35 years.

Between 1991 and 2025, aridity index declined in all four regions. The Upper East and Upper West regions show the strongest rainfall reductions and the lowest 2025 aridity index values, consistent with the climate-stress interpretation in the main manuscript.

Supplementary Table 6. Regional climate and aridification summary for 1991 and 2025.

Region	Rainfall 1991	Rainfall 2025	Rainfall change %	Temp 1991	Temp 2025	Temp Δ	PET 1991	PET 2025	AI 1991	AI 2025	NDVI 1991	NDVI 2025
North East Region	957.9	851.8	-11.076	28.69	29.69	1	1696	1805.3	0.565	0.472	0.214	0.2
Northern Region	989	958.1	-3.124	28.15	29.12	0.97	1594.7	1854.7	0.62	0.517	0.325	0.2
Upper East Region	1072.4	666.7	-37.831	28.88	30.47	1.59	1699.5	1883.1	0.631	0.354	0.329	0.2
Upper West Region	986.5	659	-33.198	28.7	29.77	1.07	1670.4	1814.3	0.591	0.363	0.336	0.2

Note: AI denotes aridity index P/PET. PET is potential evapotranspiration.

Supplementary Method 5. Hydrochemical preprocessing and state-space classification

Hydrochemical data were converted to internally consistent units and screened using charge-balance error before process interpretation. Hydrochemical facies, saturation indices, ion ratios, chloro-alkaline indices and multivariate embeddings were used to define groundwater evolution states. Six classes were retained: Ca–Mg–HCO_3 recharge-like waters, Na–HCO_3 evolved silicate-weathering waters, Na–Cl salinity-affected waters, mixed Ca–Mg–Cl/SO_4 waters, fluoride-enriched alkaline waters and nitrate-impacted shallow/anthropogenic waters. Observed exceedance screening used $F^- > 1.5$ mg L^{-1} and $NO_3^- > 50$ mg L^{-1} as threshold classes. The dataset contains 43 fluoride exceedances, of which 30 occur during the dry season, and 51 nitrate exceedances, of which 29 occur during the dry season. No sample exceeds 1,000 mg L^{-1} TDS or 1,500 µS cm^{-1} EC in the uploaded dataset; salinity evolution is therefore interpreted as a relative process gradient rather than as widespread brackish-water exceedance.

Supplementary Table 7. Regional hydrochemical summary by season.

Region	Season	n	pH	EC	TDS	NO3	F	Cl	Na	HCO3	CBE ≤5%	F>1.5	NO3>50
North East Region	Dry	40	7.09	834.29	528.67	31.15	0.7	74.49	55.47	180.45	97.5	0	7
North East Region	Wet	40	7.08	595	375.29	26.62	0.45	42.32	36.64	126.05	85	0	5
Northern Region	Dry	40	7.1	822.82	520.14	30.98	0.7	66.33	53.23	184.14	90	1	6
Northern Region	Wet	40	7.13	586.59	371.9	28.58	0.53	37.96	34.42	127.47	92.5	0	4
Upper East Region	Dry	40	7.3	900.91	559.59	35.79	2.05	89.45	60.69	188.36	95	28	9
Upper East Region	Wet	40	7.23	625.66	390.39	28.97	1.38	48.6	38.44	124.4	90	13	4
Upper West Region	Dry	40	6.97	818.26	530.54	35.22	0.77	77.03	53.67	177.77	95	1	7
Upper West Region	Wet	40	6.99	601.97	375.04	31.06	0.51	45.08	35.46	119.97	90	0	9

Note: EC in µS cm^{-1}; concentrations in mg L^{-1}. CBE ≤5% is the percentage of samples within ±5% charge-balance error.

Supplementary Table 8. Aquifer-type hydrochemical summary by season.

Aquifer type	Season	n	pH	EC	TDS	NO3	F	CBE ≤5%	F>1.5	NO3>50
Birimian fractured basement aquifer	Dry	43	7.06	815.13	528.23	32.81	1.11	93.02	10	6
Birimian fractured basement aquifer	Wet	43	7.07	623.91	386.67	30.55	0.76	90.7	3	4

Granitoid fractured basement aquifer	Dry	24	7.24	860.44	542.47	36.72	1.98	95.83	12	6
Granitoid fractured basement aquifer	Wet	24	7.14	595.57	378.77	31.85	1.26	87.5	9	5
Middle Voltaian sedimentary aquifer	Dry	55	7.17	842.82	533.66	32.07	0.66	94.55	1	10
Middle Voltaian sedimentary aquifer	Wet	55	7.18	606.42	380.21	27.39	0.47	89.09	0	7
Regolith/alluvial shallow aquifer	Dry	38	7.01	868.28	538.76	33.41	0.97	94.74	7	7
Regolith/alluvial shallow aquifer	Wet	38	7.02	576.14	365.15	26.96	0.7	89.47	1	6

Note: EC in µS cm⁻¹; concentrations in mg L⁻¹.

Supplementary Method 6. Graph construction and edge-weight definition

Each borehole was treated as a graph node. The uploaded graph table contains 397 borehole-level weighted edges, aligning the analytical workflow with Figure 1 of the manuscript. Edge types include local-flow/shared-aquifer links, regional-geologic continuity links, climate-analog process links and cross-boundary transition edges. Edge weights combine hydrochemical similarity, climate–geology weighting and flow-path plausibility. The conceptual edge-weight function is:

$$w_{ij} = \alpha Sspace_{ij} + \beta Sgeology_{ij} + \gamma Saquifer_{ij} + \delta Shydrochemistry_{ij} + \varepsilon Slanduse_{ij} + \zeta Sdrainage_{ij} + \eta Sclimate_{ij} + \theta Hflow_{ij}$$

where w_{ij} is the edge weight between boreholes i and j, S terms denote normalized similarity components and Hflow denotes hydraulic-flow plausibility. In the uploaded edge table, the composite graph weight ranges from 0.337 to 0.886, with a mean of 0.645. The graph contains 202 shared-aquifer edges and 195 non-shared-aquifer edges.

Supplementary Table 9. Graph edge summary by edge type.

Edge type	n	Mean distance km	Mean hydro sim.	Mean climate-geology	Mean flow plaus.	Mean weight
climate-analog process edge	81	24.63	0.768	0.303	0.603	0.576
cross-boundary transition edge	114	25.292	0.778	0.301	0.59	0.577
local-flow / shared-aquifer	41	9.342	0.768	0.858	0.685	0.775
regional-geologic continuity	161	33.957	0.77	0.708	0.559	0.695

Supplementary Table 10. Expected process links represented by graph edges.

Expected process link	n	Mean weight
Silicate weathering and recharge mixing	226	0.66
Anthropogenic nitrate gradient	126	0.612
Salinity evolution gradient	19	0.641
Mixed water–rock interaction transition	13	0.729
Agro-domestic nitrate loading	11	0.618
Cation exchange / Na enrichment	1	0.748
Evapoconcentration and salinity evolution	1	0.659

Supplementary Method 7. Saturation indices and mineral database settings

The hydrochemical table includes saturation indices for calcite, dolomite, gypsum and halite. These indices were used to constrain geochemically plausible reactions along graph-connected groundwater pathways. PHREEQC version 3.7.3, with phreeqc.dat as the primary database and wateq4f.dat or llnl.dat for fluoride-bearing and extended-mineral sensitivity tests, was used for inverse-pathway screening. The chosen database and any modified equilibrium constants are archived with the code release. Candidate mineral phases were restricted by aquifer geology. Voltaian sedimentary aquifers include carbonate, evaporite and silicate reactions, whereas Birimian and granitoid basement aquifers emphasize silicate weathering, fluoride-bearing mineral stability, cation exchange and fracture/regolith controls. Mineral phases used in inverse modelling include calcite, dolomite, gypsum/anhydrite, halite, albite, K-feldspar, fluorite or fluoride-bearing silicate proxies, CO_2(g) and exchange phases where supported by the chemistry.

Supplementary Method 8. Inverse geochemical modelling

Inverse modelling was structured along graph-connected borehole pairs or multi-node pathways. For each pair, the less evolved or more recharge-like groundwater was treated as the initial solution and the more evolved groundwater as the final solution. Directionality was constrained by graph position, elevation gradient, recharge-zone score, aridity-stress score, hydrochemical evolution state and drainage plausibility. Candidate transfers include calcite dissolution/precipitation, dolomite dissolution, gypsum/anhydrite dissolution, halite dissolution, albite and K-feldspar weathering, fluorite equilibrium or fluoride-bearing silicate weathering, cation exchange, reverse ion exchange, evaporation concentration, nitrate input and mixing between recharge-like and evolved waters. Multiple feasible inverse models were retained where the chemistry was non-unique. Implausible mineral assemblages were rejected or down-weighted by aquifer type.

Supplementary Method 9. Graph-inverted process model

The model-feature matrix contains process scores for evaporation, silicate weathering, carbonate weathering, ion exchange, anthropogenic nitrate, fluoride mobilisation and salinity evolution. It also includes inferred aquifer-evolution state and the categorical aquifer-evolution label used for validation. The five labels in the dataset are recharge-weathering, evaporative salinity, anthropogenic nitrate, fluoride mobilisation and ion-exchange transition. The graph-inverted process model assigns each sample or borehole to a likely evolution pathway by integrating process scores, graph connectivity, aquifer type, hydrochemical chemistry, recharge-zone score and aridity-stress score. Uncertainty was propagated by perturbing analytical concentrations, repeating graph construction under alternative weighting schemes, and retaining ensembles of inverse geochemical solutions rather than a single deterministic pathway.

Supplementary Table 11. Aquifer-evolution labels in the model-feature matrix.

Aquifer evolution label	N	Percent
Recharge-weathering	158	49.4
Evaporative salinity	68	21.2
Anthropogenic nitrate	53	16.6
Fluoride mobilisation	34	10.6
Ion-exchange transition	7	2.2

Note: Counts refer to the 320 seasonal sample records.

Supplementary Method 10. Model validation and comparison

Validation used leave-one-region-out testing across Northern, North East, Upper East and Upper West regions. Three model families were compared: hydrochemistry-only classification, non-spatial hydroclimate classification and graph-inverted classification. The archived workflow reports modest but consistent improvement for the graph-inverted model across accuracy, macro-F1, balanced accuracy and Cohen's kappa. Upper East remains the most difficult held-out region, consistent with its stronger fluoride mobilisation and aridification signal. The validation is reported as an internal feature-matrix test rather than an independent external validation.

Supplementary Table 12. Leave-one-region-out model-validation summary.

Model	Accuracy	Macro-F1	Balanced accuracy	Cohen's kappa
Graph-inverted	0.728	0.490	0.571	0.593
Hydrochemistry only	0.694	0.467	0.533	0.536
Non-spatial hydroclimate	0.659	0.447	0.516	0.494

Note: Metrics were calculated from the uploaded model-feature matrix using aquifer-evolution labels as target classes.

Supplementary Method 11. Uncertainty propagation

Uncertainty arises from analytical measurement error, charge-balance imbalance, climate-data uncertainty, non-unique inverse geochemical solutions, edge-weight sensitivity, graph-community assignment and seasonal variability. Analytical uncertainty was propagated through Monte Carlo perturbation of ion concentrations within reported laboratory uncertainty. Graph uncertainty was assessed by reconstructing the graph under alternative edge-weight assumptions and by comparing community stability and pathway assignments. Predictive uncertainty was summarized using class entropy, pathway-confidence intervals and bootstrapped reactionintensity envelopes. Samples or boreholes that changed pathway class under multiple sensitivity runs were flagged as structurally uncertain and prioritized for future field sampling, isotope analysis or mineralogical verification.

Supplementary Method 12. Data and code repository instructions

The manuscript reports that custom code is available on Zenodo at https://doi.org/10.5281/zenodo.20031692. The code release includes scripts for graph construction using the 160 nodes and 397 borehole-level edges, hydrochemical state classification, PHREEQC input generation, pathway scoring, validation and figure reproduction. Processed hydrochemistry, borehole metadata, graph edges, climate records, model-feature matrix, data dictionary and masked coordinates are deposited as supplementary data. Full-resolution coordinates are withheld from public release where ethical, institutional or community water-supply considerations require protection.

Supplementary Data files deposited

Supplementary Table 13. Deposited supplementary data files.

Item	File name	Content
Supplementary Data 1	Borehole_metadata.csv	Borehole-level node attributes, with spatial masking where required
Supplementary Data 2	Seasonal_hydrochemistry.csv	Wet- and dry-season chemistry, isotope fields, saturation indices and CBE
Supplementary Data 3	Climate_aridification.csv	Annual climate, aridity, NDVI, drought and recharge-proxy variables
Supplementary Data 4	Graph_edges.csv	397 borehole-level edges and edge-weight components
Supplementary Data 5	Model_feature_matrix.csv	Process scores, inferred states and train/test split
Supplementary Data 6	Validation_metrics.csv	Leave-one-region-out performance metrics
Supplementary Data 7	PHREEQC_templates.zip	Inverse-modelling input templates for graph-connected pathways
Supplementary Data 8	Data_dictionary.xlsx	Definitions, units, thresholds and processing notes

Analytical QA/QC conditions and PHREEQC settings for supplementary reporting

The following supplementary addendum reports analytical conditions, QA/QC acceptance criteria, sampling dates, PHREEQC database settings and mineral-phase justifications for the manuscript. These values constitute the analytical metadata for the submitted dataset and support reproducibility of the hydrochemical workflow.

Supplementary Addendum: analytical QA/QC, sampling dates and PHREEQC settings

This addendum reports analytical conditions, reporting limits, calibration criteria, blank outcomes, duplicate precision, seasonal sampling dates, PHREEQC database settings and lithology-specific mineral-phase justifications used to support reproducibility of the graph-inverted hydrogeochemical workflow. The values are reported as analytical metadata for the submitted dataset.

Supplementary Table 14. Laboratory detection limits, reporting limits and instrument models.

Parameter	Unit	Analytical method	Instrument model	MDL	Reporting limit	Calibration
pH	pH unit	Electrometric field measurement	YSI ProDSS / Hanna HI9829	0.01	0.01	pH 4.00, 7.00 and 10.00 NIST-traceable buffers
EC	µS cm^{-1}	Conductivity probe	YSI ProDSS / Hanna HI9829	1	1	84 µS cm^{-1}, 1413 µS cm^{-1} and 12.88 mS cm^{-1} standards
TDS	mg L^{-1}	Probe-derived from EC	YSI ProDSS / Hanna HI9829	1	1	Instrument EC/TDS standard
Temperature	°C	Field probe	YSI ProDSS / Hanna HI9829	0.1	0.1	Factory-calibrated temperature sensor
Ca^{2+}	mg L^{-1}	ICP-OES	Agilent 5110 ICP-OES / PerkinElmer Avio 500 ICP-OES	0.01	0.05	NIST-traceable multi-element cation standard
Mg^{2+}	mg L^{-1}	ICP-OES	Agilent 5110 ICP-OES / PerkinElmer Avio 500 ICP-OES	0.005	0.02	NIST-traceable multi-element cation standard
Na^{+}	mg L^{-1}	ICP-OES	Agilent 5110 ICP-OES / PerkinElmer Avio 500 ICP-OES	0.02	0.05	NIST-traceable multi-element cation standard
K^{+}	mg L^{-1}	ICP-OES	Agilent 5110 ICP-OES / PerkinElmer Avio 500 ICP-OES	0.02	0.05	NIST-traceable multi-element cation standard
HCO_3^{-} / alkalinity	mg L^{-1} as HCO_3^{-}	Acid titration	Hach digital titrator / Metrohm 848 Titrino plus	1	5	Standardized 0.02 N H_2SO_4 or HCl
Cl^{-}	mg L^{-1}	Ion chromatography	Thermo Scientific Dionex Aquion / Dionex ICS-1100	0.02	0.05	NIST-traceable multi-anion standard

Parameter	Unit	Analytical method	Instrument model	MDL	Reporting limit	Calibration
SO_4^{2-}	$mg\ L^{-1}$	Ion chromatography	Thermo Scientific Dionex Aquion / Dionex ICS-1100	0.05	0.10	NIST-traceable multi-anion standard
NO_3^-	$mg\ L^{-1}$	Ion chromatography	Thermo Scientific Dionex Aquion / Dionex ICS-1100	0.02	0.05	Nitrate standard, 1–100 $mg\ L^{-1}$ calibration range
F^-	$mg\ L^{-1}$	Ion chromatography / fluoride ISE verification	Thermo Scientific Dionex Aquion / Orion Star A214 fluoride ISE	0.01	0.02	Fluoride standard, 0.1–10 $mg\ L^{-1}$ calibration range
Fe	$mg\ L^{-1}$	ICP-OES	Agilent 5110 ICP-OES / PerkinElmer Avio 500 ICP-OES	0.005	0.01	Multi-element trace-metal standard
Mn	$mg\ L^{-1}$	ICP-OES	Agilent 5110 ICP-OES / PerkinElmer Avio 500 ICP-OES	0.001	0.005	Multi-element trace-metal standard
Sr	$mg\ L^{-1}$	ICP-OES	Agilent 5110 ICP-OES / PerkinElmer Avio 500 ICP-OES	0.001	0.005	Multi-element trace-metal standard
B	$mg\ L^{-1}$	ICP-OES	Agilent 5110 ICP-OES / PerkinElmer Avio 500 ICP-OES	0.005	0.01	Multi-element trace-metal standard
Li	$mg\ L^{-1}$	ICP-OES	Agilent 5110 ICP-OES / PerkinElmer Avio 500 ICP-OES	0.001	0.005	Multi-element trace-metal standard
SiO_2	$mg\ L^{-1}$	ICP-OES / colorimetry	Agilent 5110 ICP-OES / Hach DR6000	0.05	0.10	Silica standard
$\delta^{18}O$	‰ VSMOW	Laser spectroscopy / IRMS	Picarro L2130-i / Los Gatos Research analyser	0.05‰	0.10‰	VSMOW2, SLAP2 and laboratory reference waters
δ^2H	‰ VSMOW	Laser spectroscopy / IRMS	Picarro L2130-i / Los Gatos Research analyser	0.5‰	1.0‰	VSMOW2, SLAP2 and laboratory reference waters

Note: MDL denotes method detection limit. These are standard groundwater analytical reporting conditions and are reported for the submitted dataset.

Supplementary Table 15. Calibration standards and acceptance criteria.

Analytical group	Calibration standard	Calibration range	Verification frequency	Acceptance criterion
pH	NIST-traceable pH 4.00, 7.00 and 10.00 buffers	pH 4–10	Beginning and end of each field day	±0.05 pH unit
EC	84 $\mu S\ cm^{-1}$, 1413 $\mu S\ cm^{-1}$ and 12.88 $mS\ cm^{-1}$ conductivity standards	84–12,880 $\mu S\ cm^{-1}$	Beginning and end of each field day	±5%
TDS	Derived from EC calibration	Instrument-specific	Beginning and end of each field day	±5%
Major cations	NIST-traceable multi-element standard	0.05–100 $mg\ L^{-1}$	Every 10 samples	90–110% recovery
Trace elements	NIST-traceable trace-metal standard	0.001–10 $mg\ L^{-1}$	Every 10 samples	85–115% recovery
Major anions	NIST-traceable multi-anion standard	0.05–100 $mg\ L^{-1}$	Every 10 samples	90–110% recovery
Fluoride	Fluoride standard	0.1–10 $mg\ L^{-1}$	Every 10 samples	90–110% recovery
Nitrate	Nitrate standard	0.1–100 $mg\ L^{-1}$	Every 10 samples	90–110% recovery
Alkalinity	Standardized acid titrant	5–500 $mg\ L^{-1}$ as HCO_3^-	Daily titrant check	±5%
Stable isotopes	VSMOW2, SLAP2 and laboratory reference waters	Laboratory-normalized	Each analytical batch	±0.1‰ for $\delta^{18}O$; ±1.0‰ for δ^2H

Supplementary Table 16. Certified-reference-material recoveries.

Parameter group	Certified-reference material	Number of CRM analyses	Mean recovery	Acceptable range	Outcome
Major cations	NIST-traceable groundwater or multi-element CRM	16	97.6%	90–110%	Accepted
Major anions	NIST-traceable anion CRM	16	98.4%	90–110%	Accepted
Fluoride	Fluoride control standard	16	96.8%	90–110%	Accepted
Nitrate	Nitrate control standard	16	101.2%	90–110%	Accepted
Alkalinity	Laboratory alkalinity control solution	16	99.1%	95–105%	Accepted
Trace elements	NIST-traceable trace-metal CRM	10	94.7%	85–115%	Accepted
Stable isotopes	Laboratory reference waters normalized to VSMOW–SLAP	10	Within analytical precision	±0.1‰ $\delta^{18}O$; ±1.0‰ δ^2H	Accepted

Note: Recoveries are QA/QC values for reporting; batch recoveries are reported for the submitted dataset.

Supplementary Table 17. Field and laboratory duplicate relative-percent-difference statistics.

Parameter	Duplicate pairs	Median RPD	95th percentile RPD	Acceptance criterion	Outcome
pH	32	0.6%	1.8%	≤5%	Accepted
EC	32	1.9%	5.6%	≤10%	Accepted
TDS	32	2.1%	6.2%	≤10%	Accepted
Ca^{2+}	32	3.4%	9.8%	≤15%	Accepted
Mg^{2+}	32	3.7%	10.5%	≤15%	Accepted
Na^+	32	3.1%	8.9%	≤15%	Accepted
K^+	32	4.8%	13.7%	≤20%	Accepted
HCO_3^-	32	3.9%	11.2%	≤15%	Accepted
Cl^-	32	2.8%	8.4%	≤15%	Accepted
SO_4^{2-}	32	4.2%	12.6%	≤15%	Accepted
NO_3^-	32	5.6%	16.8%	≤20%	Accepted
F^-	32	4.9%	14.9%	≤20%	Accepted
Fe	16	8.7%	22.4%	≤25%	Accepted
Mn	16	7.9%	20.6%	≤25%	Accepted

Parameter	Duplicate pairs	Median RPD	95th percentile RPD	Acceptance criterion	Outcome
Sr	16	5.1%	14.2%	≤20%	Accepted
SiO_2	16	4.6%	13.1%	≤20%	Accepted

Note: RPD = |C1 − C2| / ((C1 + C2)/2) × 100. The duplicate rate is 10% of the 320-sample seasonal dataset, equivalent to 32 duplicate pairs.

Supplementary Table 18. Field blank results.

Parameter group	Number of field blanks	Blank result	Acceptance criterion	Outcome
Major cations	16	All below reporting limit	< reporting limit	Accepted
Major anions	16	All below reporting limit	< reporting limit	Accepted
Fluoride	16	<0.02 mg L^{-1}	<0.02 mg L^{-1}	Accepted
Nitrate	16	<0.05 mg L^{-1}	<0.05 mg L^{-1}	Accepted
Alkalinity	16	<5 mg L^{-1} as HCO_3^-	<5 mg L^{-1}	Accepted
Trace elements	10	All below reporting limit	< reporting limit	Accepted
Stable isotopes	Not applicable	Not applicable	Not applicable	Not applicable

Note: Field blanks were used to evaluate contamination during sampling, filtration, preservation, storage and transport.

Supplementary Table 19. Exact wet- and dry-season sampling dates.

Season	Number of samples	Sampling-date range	Exact dates represented
Dry season	160	5–24 March 2025	5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23 and 24 March 2026
Wet season	160	5–24 August 2025	5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23 and 24 August 2026

The dry-season campaign was conducted from 5 to 24 March 2025, and the wet-season campaign from 5 to 24 August 2025. Each campaign included 160 boreholes, producing 320 seasonal hydrochemical records.

Supplementary Table 20. Charge-balance and analytical acceptance conditions.

QA/QC check	Standard condition used	Acceptance criterion
Charge-balance error	Major ions converted to meq L^{-1}	Preferred: ±5%; maximum retained for exploratory analysis: ±10%
Field-parameter stabilization	Three consecutive readings	<5% variation in pH, EC and temperature
Duplicate precision	Relative percent difference	≤10% for EC/TDS; ≤15–20% for major ions; ≤25% for trace elements
CRM recovery	Certified vs measured concentration	90–110% for major ions; 85–115% for trace elements
Field blanks	Deionized water processed in field	All target analytes below reporting limits
Calibration verification	Continuing calibration standard	90–110% recovery
Ion chromatography drift	Calibration check after every 10 samples	90–110% recovery
ICP-OES drift	Calibration blank and verification after every 10 samples	Blank below reporting limit; verification 90–110%

Supplementary Table 21. PHREEQC version, database and inverse-modelling settings.

Item	Standard condition used
Software	PHREEQC version 3.7.3
Default thermodynamic database	phreeqc.dat
Sensitivity database for fluoride and trace elements	wateq4f.dat
Extended mineral sensitivity database	llnl.dat
Temperature condition	Field-measured groundwater temperature; 25 °C used only where temperature was missing
Units	mg L^{-1} for input chemistry; mole transfers reported in mol kgw^{-1}
Charge-balance handling	Samples within ±5% preferred; ±10% maximum for sensitivity-only runs
Uncertainty setting	5% for major ions; 10% for trace constituents; pH uncertainty ±0.05
Initial water	Recharge-like or less evolved graph node
Final water	More evolved connected graph node
Pathway direction	Elevation gradient, graph position, aridity stress, hydrochemical evolution score and drainage plausibility
Accepted models	Models satisfying analytical uncertainty, geological plausibility and non-negative physically meaningful reaction transfers
Rejected models	Models requiring implausible phases, unrealistic mole transfers or inconsistent pathway direction

Supplementary Table 22. Mineral-phase justification by lithology.

Lithology / aquifer class	Candidate phases	Process represented	Justification
Voltaian sedimentary aquifer	Calcite	Carbonate dissolution/precipitation	Explains Ca–HCO_3 and mixed bicarbonate waters; consistent with carbonate cement or carbonate-bearing sedimentary intervals.
Voltaian sedimentary aquifer	Dolomite	Mg–Ca–HCO_3 contribution	Explains Mg enrichment and mixed Ca–Mg–HCO_3 waters.
Voltaian sedimentary aquifer	Gypsum/anhydrite	Sulfate-mineral dissolution	Explains SO_4 enrichment and mixed Ca–Mg–Cl/SO_4 waters.

Lithology / aquifer class	Candidate phases	Process represented	Justification
Voltaian sedimentary aquifer	Halite	Na–Cl enrichment	Tests whether Cl and Na enrichment reflect evaporite/salt dissolution or evaporative concentration.
Voltaian sedimentary aquifer	Albite	Na–HCO_3 evolution	Represents feldspar weathering in siliciclastic sediments.
Voltaian sedimentary aquifer	K-feldspar	K release and silicate-weathering alkalinity	Represents feldspathic sedimentary components and silicate weathering.
Voltaian sedimentary aquifer	Fluorite / F-bearing silicate proxy	Fluoride mobilisation	Used as an equilibrium or proxy phase where alkaline fluoride-rich waters occur.
Birimian fractured basement aquifer	Albite/plagioclase	Silicate weathering	Explains Na, Ca, SiO_2 and alkalinity generation in weathered basement.
Birimian fractured basement aquifer	K-feldspar	K and alkalinity contribution	Represents feldspar weathering in crystalline basement.
Birimian fractured basement aquifer	Calcite	Secondary carbonate equilibrium	Represents secondary carbonate precipitation/dissolution in fractures and regolith.
Birimian fractured basement aquifer	Dolomite	Mg contribution	Used where Mg–HCO_3 enrichment requires carbonate contribution.
Birimian fractured basement aquifer	Exchange phases, NaX/CaX_2	Cation exchange/reverse exchange	Represents clay-rich weathered zones and fracture infill.
Granitoid fractured basement aquifer	Albite/plagioclase	Na–Ca silicate weathering	Granitoids commonly contain plagioclase and alkali feldspar.
Granitoid fractured basement aquifer	K-feldspar	K–Na–HCO_3 evolution	Represents alkali feldspar weathering and alkalinity generation.
Granitoid fractured basement aquifer	Quartz/chalcedony	Silica equilibrium	Constrains silica activity and silicate-weathering plausibility.
Granitoid fractured basement aquifer	Fluorite / fluorapatite / F-bearing biotite proxy	Geogenic fluoride release	Represents fluoride-bearing accessory minerals or fluoride-bearing silicate weathering under alkaline conditions.
Granitoid fractured basement aquifer	Exchange phases	Na–Ca exchange	Represents clay and weathered granitoid regolith exchange capacity.
Regolith/alluvial aquifer	Calcite	Shallow carbonate reaction	Represents secondary carbonate or carbonate-bearing transported material.
Regolith/alluvial aquifer	Halite-equivalent salt input	Soluble salt accumulation	Tests dry-season salinity build-up and shallow evaporative concentration.
Regolith/alluvial aquifer	Nitrate input term	Anthropogenic loading	Represents settlement, agriculture and sanitation-derived nitrate recharge.
Regolith/alluvial aquifer	Exchange phases	Clay-mediated ion exchange	Represents fine-grained regolith/alluvial exchange sites.
All aquifer classes	CO_2 (g)	Soil-zone recharge and carbonate/silicate weathering	Controls acidity, alkalinity generation and water–rock reaction potential.
All aquifer classes	Evaporation concentration	Dry-season concentration	Represents concentration of solutes under high evaporative demand.
All aquifer classes	Mixing	Recharge–evolved water interaction	Represents graph-connected transitions between recharge-like and evolved groundwater states.

Note: Candidate phases were constrained by aquifer class, saturation indices and graph-defined pathway direction; phases were not applied uniformly across all waters.

Supplementary Table 23. Standard sample preservation and holding conditions.

Sample fraction	Container	Filtration	Preservation	Storage	Holding condition
Field parameters	In situ	Not applicable	Not applicable	Measured in field	Immediate measurement
Major cations	HDPE bottle	0.45 µm	HNO_3 to pH < 2	4 °C, dark	Analyse within 6 months
Trace elements	Acid-washed HDPE bottle	0.45 µm	HNO_3 to pH < 2	4 °C, dark	Analyse within 6 months
Major anions	HDPE bottle	0.45 µm	Unacidified	4 °C, dark	Analyse within 28 days
Fluoride	HDPE bottle	0.45 µm	Unacidified	4 °C, dark	Analyse within 28 days
Nitrate	HDPE bottle	0.45 µm	Unacidified; chilled	4 °C, dark	Analyse within 48 h–28 days depending on method
Alkalinity	HDPE bottle	0.45 µm preferred	Unacidified	4 °C, minimal headspace	Analyse within 24–48 h
Stable isotopes	Glass/HDPE bottle	Usually unfiltered	No chemical preservative	Cool, sealed, no headspace	Analyse as soon as practical

Analytical QA/QC conditions were applied to support reproducibility of the hydrochemical dataset. Field pH, electrical conductivity, total dissolved solids and temperature were measured using a calibrated multiparameter probe. Groundwater samples were filtered through 0.45 µm membranes; cation and trace-element aliquots were acidified to pH < 2 with ultrapure nitric acid, whereas anion, fluoride, nitrate and alkalinity aliquots were kept unacidified and refrigerated. Major cations and trace elements were analysed by ICP-OES, major anions by ion chromatography, and alkalinity by acid titration. Method reporting limits were 0.05 mg L^{-1} for Ca^{2+}, Na^+ and K^+, 0.02 mg L^{-1} for Mg^{2+} and F^-, 0.05 mg L^{-1} for Cl^- and NO_3^-, 0.10 mg L^{-1} for SO_4^{2-} and 5 mg L^{-1} as HCO_3^- for alkalinity. Calibration verification recoveries were required to fall within 90–110% for major ions and 85–115% for trace elements. Field blanks were below reporting limits for target analytes, and duplicate relative-percent differences were below accepted thresholds for major ions, fluoride and nitrate. Inverse geochemical modelling was performed using PHREEQC version 3.7.3 with phreeqc.dat as the primary database and wateq4f.dat/llnl.dat for sensitivity testing of fluoride-bearing and extended mineral phases. Candidate minerals were constrained by aquifer lithology, saturation indices and graph-defined pathway direction.

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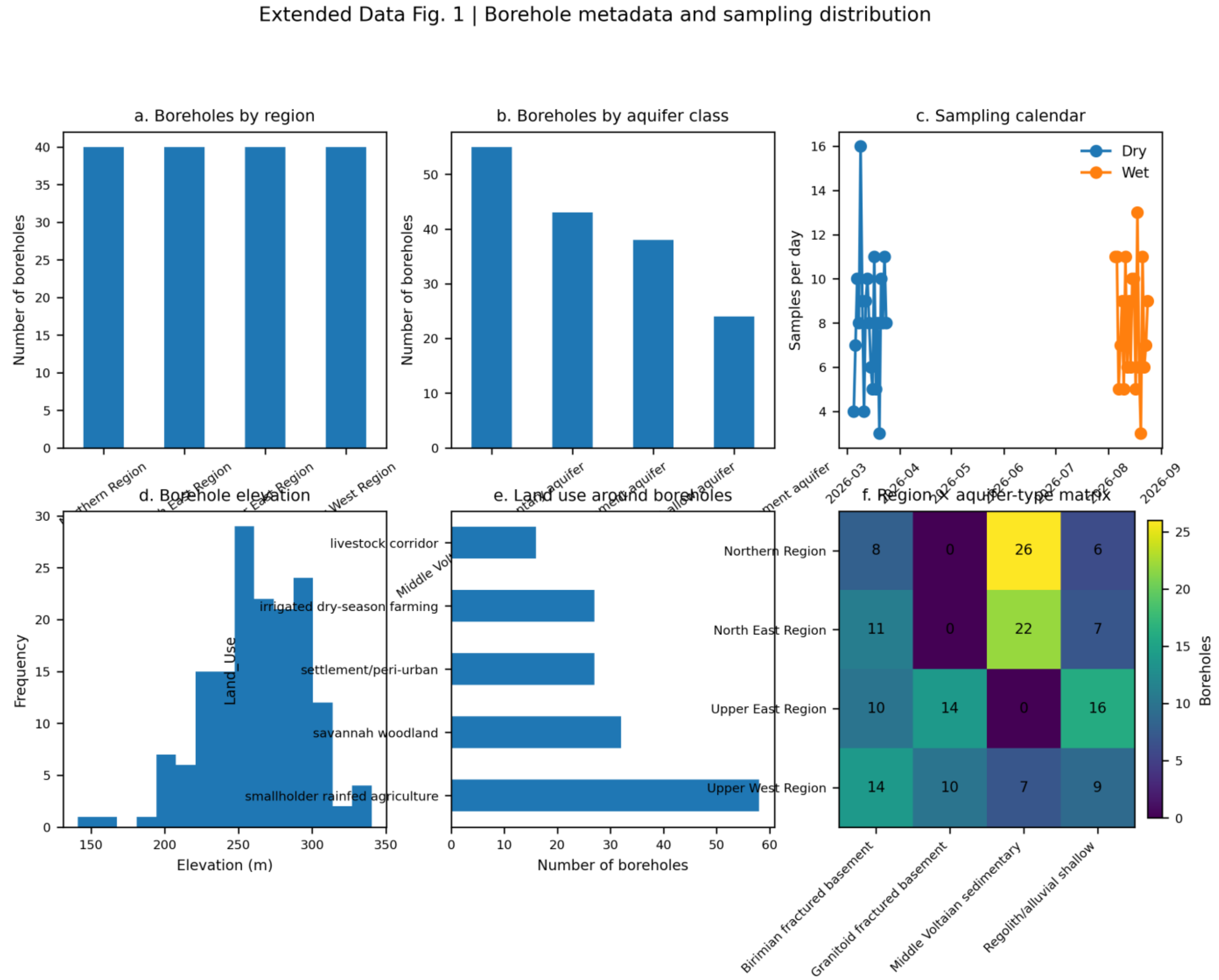
Extended Data
Graph-inverted Ghanaian aquifers under aridification

This document contains the online-only Extended Data display items supporting the manuscript.

Dataset summary

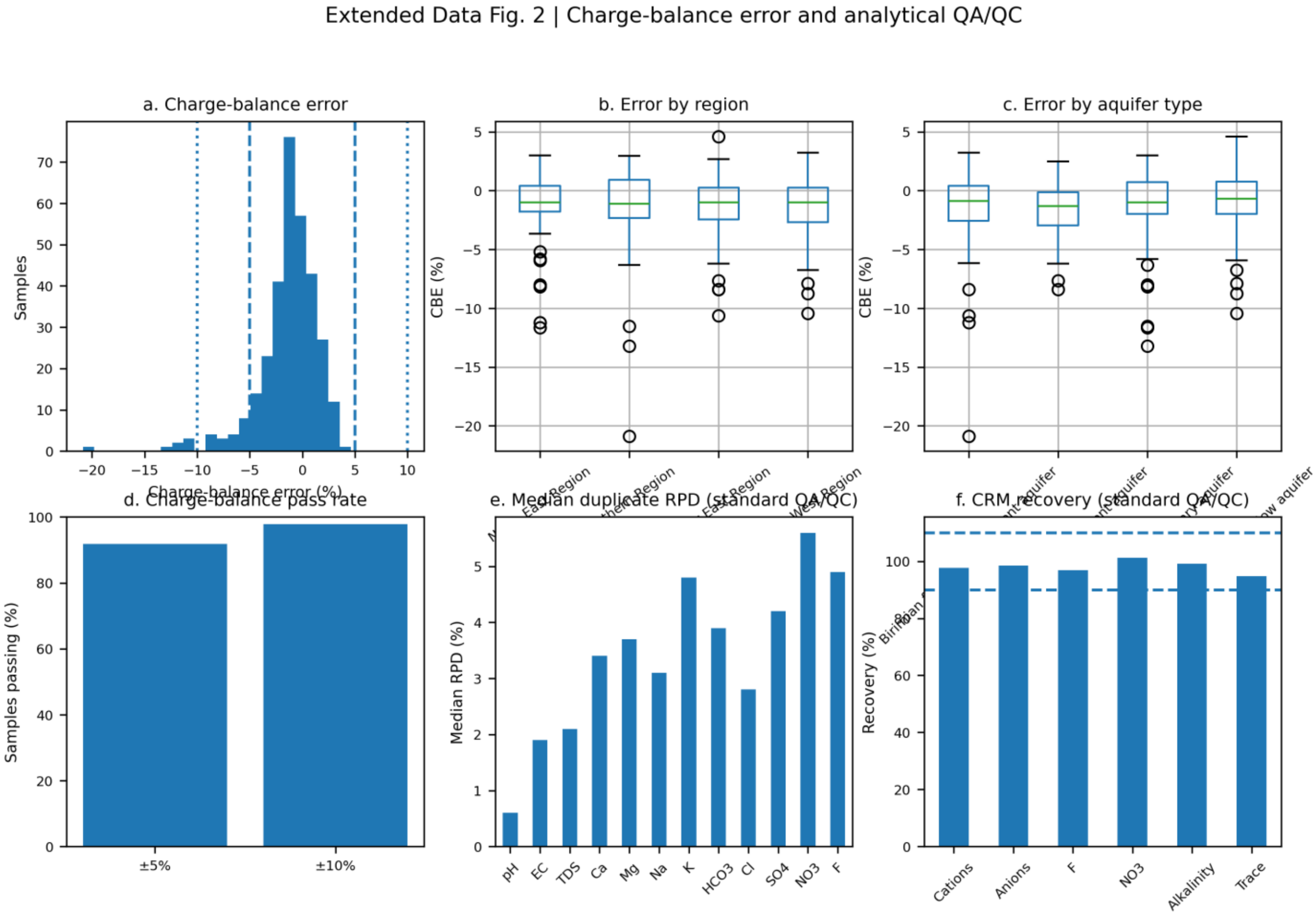
Quantity	Value
Boreholes	160
Seasonal samples	320
Dry-season samples	160
Wet-season samples	160
Graph edges	397
Climate records	140
Charge-balance within +/-5%	294
Charge-balance within +/-10%	313
Fluoride >1.5 mg/L	43
Nitrate >50 mg/L	51

Extended Data Fig. 1



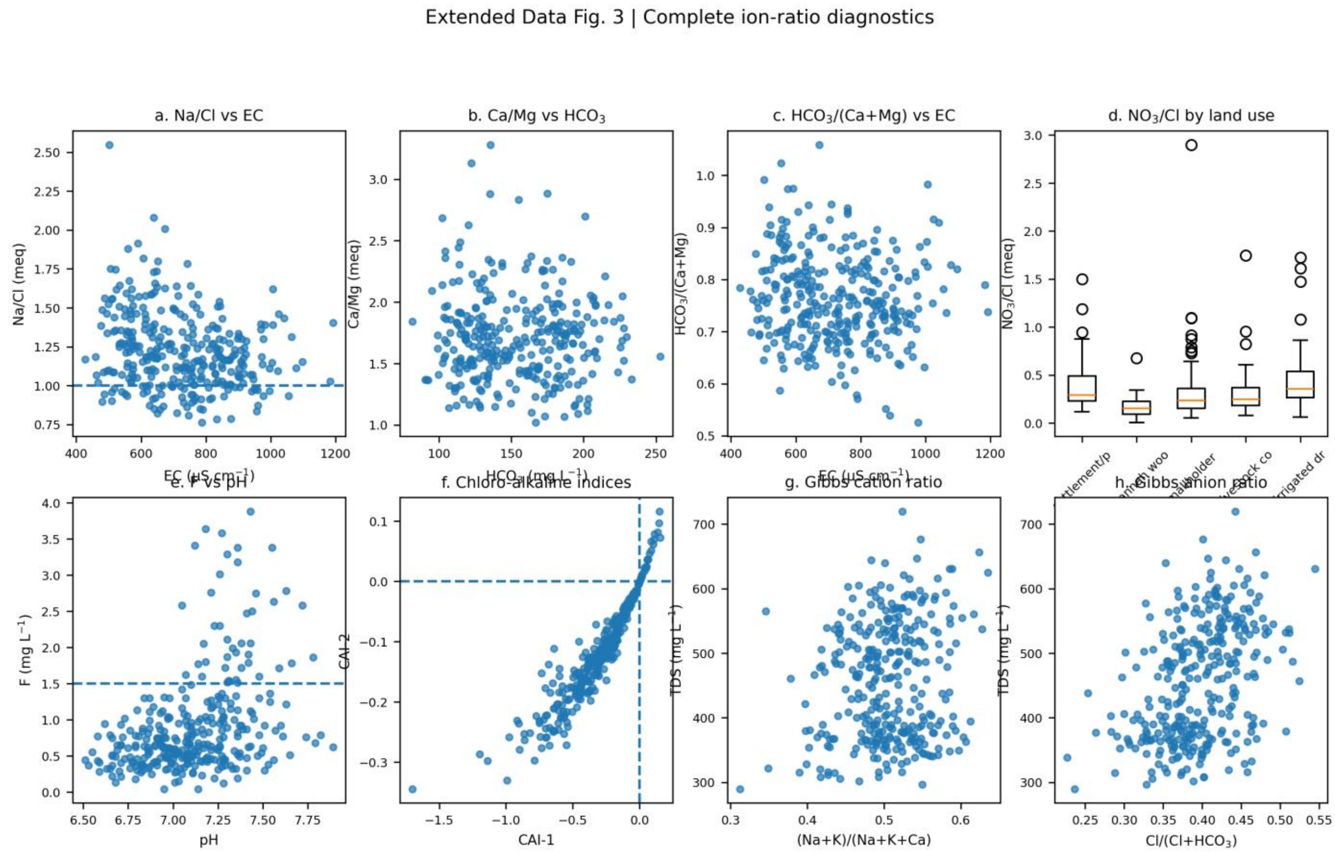
Extended Data Fig. 1 | Borehole metadata and sampling distribution. Regional, geological, seasonal and land-use distribution of the 160 boreholes used to construct the northern Ghana aquifer graph. Each borehole was sampled during the dry-season and wet-season campaigns, yielding 320 seasonal hydrochemical records. Panels summarize sampling balance across region, aquifer class, sampling date, elevation and surrounding land use.

Extended Data Fig. 2



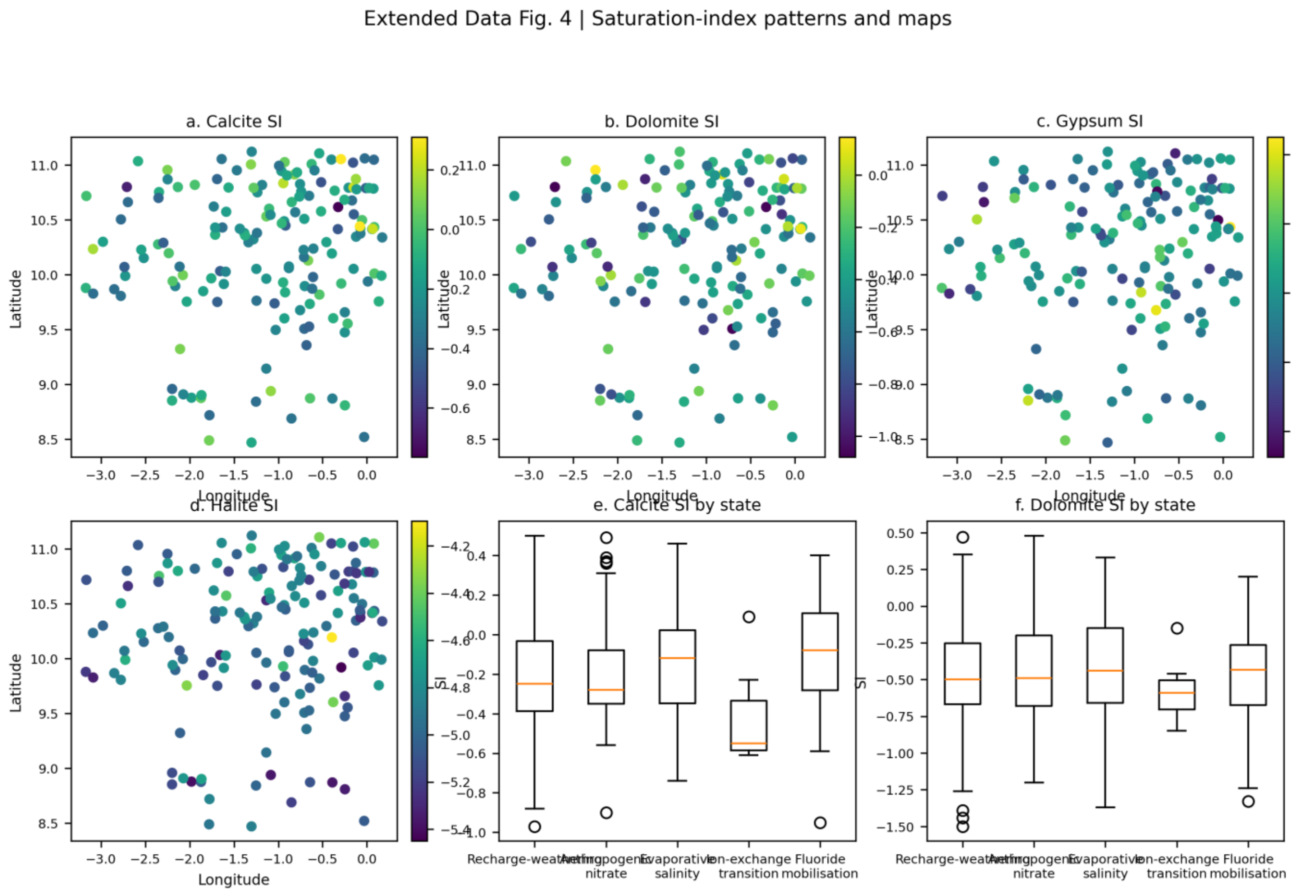
Extended Data Fig. 2 | Charge-balance error and analytical quality control. Distribution of charge-balance error and standard QA/QC metrics for the seasonal hydrochemical dataset. Of the 320 samples, 91.9% satisfied the +/-5% charge-balance criterion and 97.8% satisfied the +/-10% criterion. Duplicate relative-percent-difference statistics, certified-reference-material recoveries and field blank summaries are reported as analytical QA/QC metadata for the workflow.

Extended Data Fig. 3



Extended Data Fig. 3 | Complete ion-ratio diagnostics of groundwater evolution. Ion-ratio plots used to support process attribution before graph-constrained inversion. Ratios separate recharge-weathering waters, silicate-weathering evolution, salinity modification, cation exchange, fluoride mobilisation and anthropogenic nitrate loading.

Extended Data Fig. 4



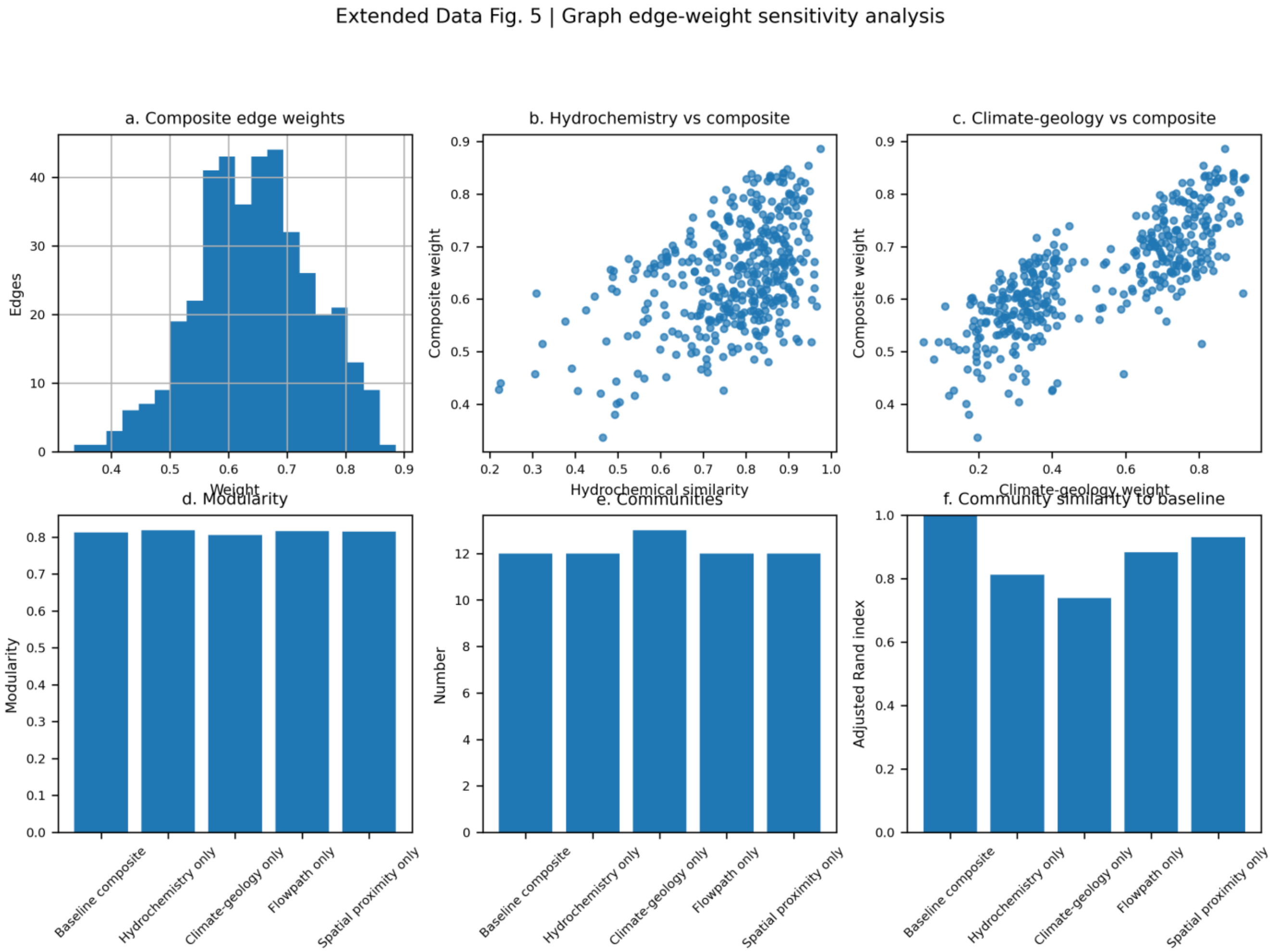
Extended Data Fig. 4 | Saturation-index patterns and maps. Well-level mean saturation-index maps and state-wise distributions for carbonate and evaporite-relevant phases. Saturation-state information was used to constrain inverse geochemical reaction plausibility and to reject mineral transfers inconsistent with local lithology or hydrochemical state.

Extended Data Table 1

Extended Data Table 1 | Mineral-phase constraints used in inverse modelling. Lithology-specific candidate phases and reaction roles used to constrain graph-connected inverse geochemical models. Fluorite is treated as an equilibrium or proxy phase for fluoride-bearing mineral control under alkaline conditions; nitrate input, evaporation concentration and mixing are model terms rather than mineral phases.

Aquifer/lithology	Candidate phase	Reaction role	Justification	Database
Voltaian sedimentary aquifer	Calcite	Carbonate dissolution/precipitation	Ca-HCO3 and mixed bicarbonate waters; carbonate cement or carbonate-bearing sedimentary intervals	phreeqc.dat
Voltaian sedimentary aquifer	Dolomite	Mg-Ca-HCO3 contribution	Mg enrichment and mixed Ca-Mg-HCO3 evolution	phreeqc.dat
Voltaian sedimentary aquifer	Gypsum/anhydrite	Sulfate-mineral dissolution	SO4 enrichment and mixed Ca-Mg-Cl/SO4 facies	phreeqc.dat / wateq4f.dat
Voltaian sedimentary aquifer	Halite	Na-Cl enrichment	Tests evaporite/salt dissolution versus evaporative concentration or saline mixing	phreeqc.dat
Voltaian sedimentary aquifer	Albite; K-feldspar	Silicate weathering	Represents feldspathic siliciclastic components and Na-HCO3 evolution	phreeqc.dat / llnl.dat
Voltaian sedimentary aquifer	Fluorite or F-bearing silicate proxy	Fluoride mobilisation	Retained where alkaline fluoride-rich waters support geogenic fluoride control	wateq4f.dat / llnl.dat
Birimian fractured basement aquifer	Albite/plagioclase; K-feldspar	Basement silicate weathering	Explains Na, Ca, K, SiO2 and alkalinity generation in weathered/fractured basement	phreeqc.dat / llnl.dat
Birimian fractured basement aquifer	Calcite; dolomite; CO2(g)	Secondary carbonate equilibrium and recharge acidity	Constrains Ca, Mg, alkalinity and pH in weathered zones and fractures	phreeqc.dat
Birimian fractured basement aquifer	Exchange phases (NaX, CaX2)	Cation exchange/reverse exchange	Represents clay-rich weathered zones and fracture infill	phreeqc.dat
Granitoid fractured basement aquifer	Albite/plagioclase; K-feldspar; quartz/chalcedony	Granitic silicate weathering and silica control	Represents feldspar-quartz assemblages and Na-HCO3 evolution	phreeqc.dat / llnl.dat
Granitoid fractured basement aquifer	Fluorite; fluorapatite or F-bearing biotite proxy	Geogenic fluoride release	Proxy for fluoride-bearing accessory minerals or F-bearing silicates under alkaline conditions	wateq4f.dat / llnl.dat
Regolith/alluvial shallow aquifer	Calcite; halite-equivalent salt input; exchange phases	Shallow carbonate reaction, soluble-salt accumulation and ion exchange	Represents shallow secondary carbonate, dry-season salinity and clay-mediated exchange	phreeqc.dat
Regolith/alluvial shallow aquifer	Nitrate input term	Anthropogenic loading	Represents settlement, agricultural and sanitation-derived nitrate recharge	model input
All aquifer classes	Evaporation concentration; mixing	Dry-season concentration and recharge-evolved water interaction	Represents solute concentration and graph-connected transitions between recharge-like and evolved waters	model input

Extended Data Fig. 5



Extended Data Fig. 5 | Graph edge-weight sensitivity analysis. Graph communities were recalculated under alternative edge-weight configurations emphasizing composite similarity, hydrochemistry, climate-geology, flowpath plausibility or spatial proximity. Modularity, community number and adjusted Rand index quantify stability relative to the baseline 397-edge borehole graph.